

Generation of sub-half-cycle ultrafast pulses in a flat-slit-confined optical field

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Compressing an ultrafast optical pulse down to a subcycle scale is an essential approach to improving temporal resolution in light-matter interaction. Considering that temporal far-field approaches have a half-cycle limit in generating subcycle pulses, recently we have proposed to break this limit in the temporal near field, and generate sub-half-cycle pulses via inverse Compton scattering in a sub-half-wavelength spatially confined optical driving field, which is generated in a line slit constructed by two sapphire hexagonal waveguides. While in principle one can generate an ultrafast optical pulse with pulse width narrower than 0.1 cycle, the practical fabrication of sapphire hexagonal waveguides and the line-slit-confinement structure is highly challenging. Here, we propose to replace the line slit with a flat slit constructed with two sapphire square waveguides, which can not only simplify the fabrication of the field-confining structure, but also evidently increase the single-pulse energy of the yielded sub-half-cycle pulse. For example, using a 0.4-THz 1-ps optical driving field and a 1-fs 3-MeV 0.1-nC electron bunch, a 0.12-cycle 5.1-fs-width midinfrared femtosecond pulse can be obtained, while the single-pulse energy is increased from 3.3 to 5.2 pJ, due to the increased fraction of the driving-field energy in the flat slit.

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I. INTRODUCTION

Ultrafast optical pulses with a timescale on the order of femtoseconds (10^{-15} s) to attoseconds (10^{-18} s) are excellent tools and platforms for investigating processes like electronic transitions and transports at the molecular and atomic scales [1,2]. Generally speaking, shorter pulse widths and higher temporal gradients enable higher temporal resolution [1], and reducing the number of pulse cycles is an essential approach to achieving shorter pulse widths. Currently, relying on various technologies such as optical parametric amplifiers [3,4], high-harmonic generation [5,6], soliton dynamics [7], coherent synthesis of optical fields [4,8,9], supercontinua [10], and relativistic electron sheets [11], pulse widths have been successfully compressed to less than a single cycle, i.e., subcycle pulses [12]. Owing to their ultrahigh temporal resolution and high electron acceleration efficiency, subcycle pulses have garnered extensive attention in multiple frontier areas of science [13], and been applied for exploring, e.g., electron tunneling time [14,15], bound electron behavior [9], dynamics of electronic band structure [16,17], ionization dynamics of molecules [18], transient absorption spectroscopy, and relativistic electron-beam generation [19].

However, similar to the well-known optical diffraction limit ($\lambda_0/2n$, where λ_0 is the vacuum wavelength of light and n is the refractive index of the medium) in the spatial far field, there is a half-cycle limit ($T_c/2$, with T_c being the optical cycle) for pulse-width compression in the temporal far field [20]. Currently, the narrowest width of a subcycle pulse in the temporal far field remains larger than $T_c/2$ [12]. Recently,

we proposed to generate sub-half-cycle ultrafast optical pulses in the temporal near field via inverse Compton scattering (ICS) between a relativistic electron bunch and sub-half-wavelength spatially confined optical fields [21,22], in which the sub-half-wavelength spatial confinement is converted into the sub-half-cycle temporal confinement [20]. With this approach, we showed the possibility of realizing a sub-half-cycle ultrafast optical pulse with a pulse width narrower than $0.1T_c$, well breaking the far-field “half-cycle limit.”

In the aforementioned approach [20], we used two coupled sapphire hexagonal waveguides to generate sub-half-wavelength spatially line-slit-confined optical fields in the terahertz (THz) band. The line-slit-confinement structure, consisting of two sapphire hexagonal waveguides featuring a submillimeter-scale cross section, requires high surface quality (to increase the damage threshold) and structural symmetry (to offer a symmetric spatial distribution of the confined optical field). Specifically, due to the small scattering cross section ($\sigma_T = 6.65 \times 10^{-29}$ m²) of the ICS, to yield an ICS sub-half-cycle pulse with energy higher than picojoule (pJ) level, the intensity of the confined optical field has to approach the material damage threshold (e.g., 2.0×10^{18} W/m² of sapphire [23]) with a currently available ultrashort electron bunch (e.g., a 0.1-nC 1-fs electron bunch [24]). To prevent the sapphire waveguide from hotspot-induced optical damage, very high structural quality (especially the surface quality) is required. Also, the construction of the line-slit structure by edge-to-edge alignment of two hexagonal waveguides requires nanometer-scale precision [21,22]; a slight defect or misalignment will distort and degrade the spatial confinement of the optical driving field [25], and consequently the temporal confinement of the ICS sub-half-cycle pulse [20].

To facilitate the experimental realization, here we propose to use sapphire square waveguides to construct a flat slit

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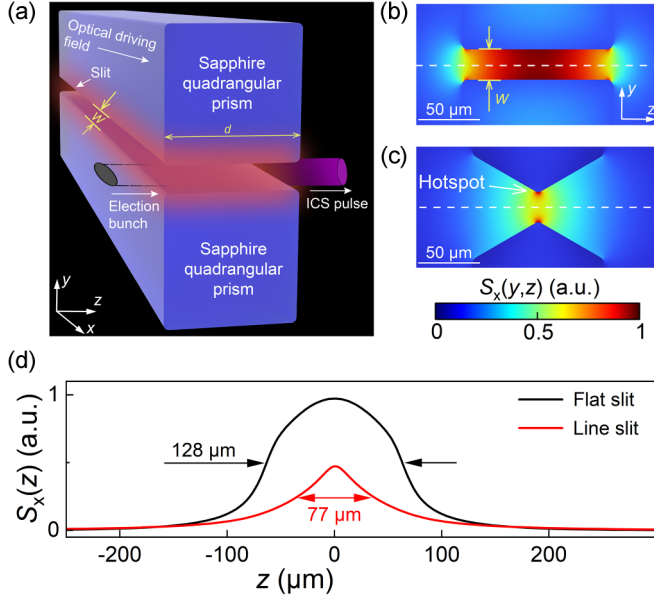


FIG. 1. Generation of sub-half-cycle pulses based on the flat-slit-confined optical field. (a) Schematic diagram of the ICS process. A relativistic electron bunch passes through the flat slit of the square waveguide from the negative z -axis direction, collides with photons confined in the slit, and generates ICS pulses. x -Poynting vector $S_x(y, z)$ of the TE_0 -like mode in the flat slit (b) and the line slit (c) under the THz driving field. (d) $S_x(z)$ along the z direction at the slit center [white dashed line in (b) and (c)] for the flat slit (black solid line) and line slit (red solid line). The flat-slit-confined optical field has $\Delta s \approx 128 \mu\text{m}$ in the electron transit direction (Δs of the line-slit-confined optical field is $\approx 77 \mu\text{m}$). Slit width W and side length d are shown in yellow arrows in (a) and (b).

instead of the line slit for optical field confinement. Compared with the hexagonal waveguide, a square waveguide is much easier to fabricate, as has been realized previously [26]. Meanwhile, the lateral face alignment (for constructing a flat slit) is also easier than the edge alignment (for constructing a line slit). Moreover, a flat slit can confine a higher fraction of field energy for ICS than a line slit, which can evidently increase the single-pulse energy of the yielded sub-half-cycle pulse.

II. CONFIGURATION OF THE FLAT-SLIT-CONFINED OPTICAL FIELD

In this approach, we place two sapphire square waveguides in parallel, and form a flat slit between the lateral faces of the two square waveguides, as shown in Fig. 1(a). In such a coupled waveguide structure, a sub-wavelength-confined field can be generated inside the slit (as part of the waveguiding mode of the coupled-waveguide structure; see Fig. 5). When a relativistic electron bunch passes through the flat slit along a direction perpendicular to the waveguide length, a sub-half-cycle pulse can be generated via ICS between the electrons and the confined optical field in the slit.

To quantify the flat-slit-confined optical field, we use a two-dimensional COMSOL simulation to calculate the x -direction (i.e., the propagation direction) Poynting vector $S_x(y,$

$z)$ of the fundamental mode (i.e., TE_0 -like mode; see Fig. 5) in the cross-sectional plane of the coupled-waveguide structure. By assuming a wavelength of the optical field to be $750 \mu\text{m}$ (i.e., 0.4 THz in frequency), a side length d of the square waveguide to be $125 \mu\text{m}$, a slit width W to be $25 \mu\text{m}$, and a refractive index of sapphire to be 3.08 at $750\text{-}\mu\text{m}$ wavelength [27], we obtain $S_x(y, z)$ shown in Fig. 1(b). For comparison, $S_x(y, z)$ of a line-slit-confined field is also provided [Fig. 1(c)], with a diagonal diameter of $125 \mu\text{m}$ of the hexagonal waveguide. It shows that, compared with the central “hot spot” in the line slit [Fig. 1(c)], a “hot surface” with a much smaller spatial gradient is formed in the flat slit, which can hold much more field energy inside the slit for ICS. Accordingly, the spatial full width at half maximum (FWHM) (Δs) of the TE_0 -like mode along the z direction (i.e., the electron motion direction) in the flat slit ($\approx 128 \mu\text{m}$, $\approx 0.17\lambda_0$) is much larger than that in the line slit ($\approx 77 \mu\text{m}$, $\approx 0.10\lambda_0$), as shown in Fig. 1(d). It is worth mentioning that, despite the obviously increased effective path of electrons, the pulse width of the ICS pulse remains sub-half-cycle due to the sub-half-wavelength spatial confinement in the flat slit [20].

III. PRINCIPLES OF ICS PULSE GENERATION IN A CONFINED OPTICAL FIELD

For detailed calculation, we assume the optical driving field to be a pulsed TE_0 -like-mode field with a central frequency $\nu_0 = 0.4 \text{ THz}$ ($\lambda_0 = 750 \mu\text{m}$) and a pulse width $\tau_0 = 1 \text{ ps}$ (similar to that in the previous proposal), with the maximum power density (located spatially at the center of the hot surface in the flat slit and temporally at the pulse peak) equal to the damage threshold of single-crystal sapphire [see Eq. (A1)]. The electron bunch is assumed to have an elliptical cross section of $25 \mu\text{m}$ for the short axis and $125 \mu\text{m}$ for the long axis with a spatially uniform distribution, a duration of 1 fs with a temporally Gaussian distribution, an electron energy E_e of 3 MeV (i.e., $V_e = 0.99c$, where c is the speed of light in vacuum), and a charge of 0.1 nC (corresponding to an electron density of $8.6 \times 10^{23}/\text{m}^3$, the same as that in the previous work [20]).

With these assumptions, the path-dependent (i.e., z -axis) photon number density $n(z)$ of the ICS pulse is given by [28]

$$n(z) = N_0 \sigma_T \frac{\iint n_e(x, y, z) S_x(x, y, z) dx dy}{P_{\text{mode}}} \frac{\delta_T}{\tau_0}, \quad (1)$$

where $N_0 \approx E_{p0}/h\nu_0$ is the total photon number of the driving field, in which E_{p0} is the total energy of the driving pulse, σ_T is the Thomson scattering cross section, $n_e(x, y, z)$ is the electron number density of the incident relativistic electron bunch, $S_x(x, y, z)$ is the x -Poynting vector along the interaction path, P_{mode} is the total power of the TE_0 -like mode, δ_T is the electron transit time, and τ_0 is the width of the driving pulse.

Thus, the total number of photons N_{sp} of the ICS pulse can be obtained as

$$N_{\text{sp}} = \int n(z) dz. \quad (2)$$

When the electron energy is sufficiently high, the energy loss of the electron traversing the slit is negligible, and the electron trajectory is approximately a straight line (see Fig. 6).

Thus, the receiving time t of photons generated by the electron at z is given by

$$t(z, \theta) = \frac{z}{V_e}(1 - \beta \cos \theta) + \frac{r}{c}, \quad (3)$$

where $\beta = V_e/c$, θ is the scattering angle (i.e., the angle between the direction of the scattered photon and the positive z axis) with $\sin \theta \in [-1/\gamma, 1/\gamma]$ [20,28], $\gamma = (1 - \beta^2)^{-1/2}$ is the Lorentz factor, r is the radius of the spherical receiving surface centered at the center point of the slit (e.g., $r = 1 \text{ cm} \gg d$), and the factor $(1 - \beta \cos \theta)$ comes from the time compression effect for the observer facing the scattered photon [29]. The path-dependent $n(z)$ can be rewritten as the time-dependent $n(t, \theta)$:

$$n(t, \theta) = n \left[\left(t(z, \theta) - \frac{r}{c} \right) V_e / (1 - \beta \cos \theta) \right].$$

The distribution of scattered photons obeys $f(\theta, \psi) = K \chi^2 [1/2 + \chi(\chi-1)(1 + \cos 2\psi)]$ [28], where K is a constant coefficient and $\chi = 1/(1 + \gamma^2 \theta^2)$. By integrating over the azimuthal angle ψ , the intensity of the scattered pulse on the receiving surface, $I(t, \theta)$, can be obtained as

$$I(t, \theta) = n(t, \theta) f(\theta) h \nu(\theta), \quad (4)$$

where t is a function of θ [Eq. (3)]; $h = 6.626 \times 10^{-34} \text{ J s}$ is the Planck constant. When the electron incidence direction is perpendicular to the wave propagation direction, the frequency of high-energy photons $\nu(\theta)$ generated by ICS is given by [28]

$$\nu(\theta) = \frac{\nu_0}{1 - \beta \cos \theta}.$$

Since the phase of the scattered field is inherited from the optical driving field [30] and the electron bunch duration is extremely short compared to the driving pulse width, we assume that the amplitude of the driving field reaches its positive maximum when the electron arrives at the center point of the slit (i.e., $z = 0$). In addition, since the cycle of the driving pulse is much larger than the time of the electrons transiting through the sub-half-wavelength confined driving field (2.5 vs 0.43 ps), a slight misalignment (e.g., 1 ps) in the arrival time between the electron bunch and the driving pulse peak will not significantly affect the symmetry and intensity of the ICS pulse. Due to the linear polarization (y polarization; see Fig. 5) of the flat-slit-confined optical field, the electric field of the ICS pulse $E(t, \theta)$ only has the $E_y(t, \theta)$ component. Thus, in the spatial far field, $E_y(t, \theta)$ can be expressed as

$$E_y(t, \theta) = \sqrt{2} \sqrt{\frac{\mu_0}{\varepsilon_0}} I(t, \theta) e^{i2\pi \nu_0 \left(\frac{t}{1 - \beta \cos \theta} \right)}, \quad (5)$$

where μ_0 and ε_0 are the vacuum permeability and vacuum permittivity, respectively. By integrating Eqs. (4) and (5) over the scattering angle θ , time-dependent intensity $I(t)$ and electric field $E_y(t)$ of the ICS pulse can be obtained as

$$I(t) = \int_{-\arcsin(-1/\gamma)}^{\arcsin(-1/\gamma)} I(t, \theta) d\theta, \quad (6)$$

$$E_y(t) = \int_{-\arcsin(-1/\gamma)}^{\arcsin(-1/\gamma)} E_y(t, \theta) d\theta. \quad (7)$$

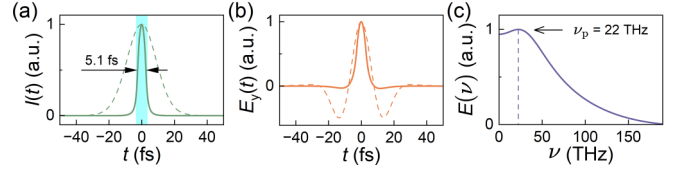


FIG. 2. Sub-half-cycle femtosecond ICS pulse generated with a 0.4-THz 1-ps flat-slit-confined optical field (with Δs of 128 μm) and a 3-MeV 1-fs electron bunch. (a) Intensity $I(t)$ and (b) electric field $E_y(t)$ of an ICS pulse generated in a flat-slit-confined optical field. For comparison, $I(t)$ and $E_y(t)$ of an ICS pulse generated in a 0.4-THz spatially Gaussian-profile optical driving field are also provided by the dashed line in (a) and (b), respectively. (c) Fourier spectrum $E(\nu)$ of $E_y(t)$ of the ICS pulse, giving $\tau \approx 5.1 \text{ fs}$ and $\nu_p \approx 22 \text{ THz}$. The temporal near field is marked in the cyan area in (a).

Then, the spectrum $E(\nu)$ of the ICS pulse can be obtained by the Fourier transform of $E_y(t)$, and the total energy E_{sp} is derived by integrating the intensity $I(t)$ [Eq. (6)] over time:

$$E_{\text{sp}} = \int I(t) dt. \quad (8)$$

IV. RESULTS AND DISCUSSION

Based on the principles and parameters mentioned above, we calculated the ICS pulse generated with a spatially sub-half-wavelength-confined driving field in the flat slit, as shown in Fig. 2. Figure 2(a) gives the intensity $I(t)$ of the ICS pulse obtained from Eq. (6), with a pulse width $\tau \approx 5.1 \text{ fs}$. Similar to the spatial near field, the temporal near field—confined around the origin of the temporal coordinate (i.e., $t = 0$) with a higher time gradient (i.e., larger dI/dt) than that generated with a spatially Gaussian-profile optical driving field [dashed lines in Figs. 2(a) and 2(b)]—is also shown as the cyan region. Figure 2(b) shows the electric field $E_y(t)$ of the ICS pulse calculated via Eq. (7), showing an obvious unipolar feature [31,32] [i.e., $\int E_y(t) dt \neq 0$]. Figure 2(c) shows the Fourier spectrum $E(\nu)$ of $E_y(t)$, which has a nonzero dc component (a typical feature of a unipolar pulse) and a peak frequency of $\nu_p \approx 22 \text{ THz}$ (corresponding to $T_c \approx 44 \text{ fs}$ and a peak wavelength $\lambda_p \approx 13 \mu\text{m}$). Consistent with previous work [20], we define temporal confinement factor $\zeta = T_c/\tau$ to quantitatively describe the degree of temporal confinement of ultrafast pulses (e.g., $\zeta = 2$ denotes a half-cycle pulse). The ζ value of the aforementioned sub-half-cycle pulse is ≈ 8.7 , corresponding to a 0.12-cycle 5.1-fs-width midinfrared femtosecond pulse.

Meanwhile, the single-pulse photon number (N_{sp}) of the ICS pulse obtained from Eq. (2) is $\approx 1.4 \times 10^8$, with a single-pulse energy (E_{sp}) derived from Eq. (8) of $\approx 5.2 \text{ pJ}$ (higher than the minimum detectable energy of $\approx 1 \text{ pJ}$ of early attosecond pulses [33]). Compared with those in the previous work [20], the N_{sp} and E_{sp} of the ICS pulse are increased by ≈ 2.0 times (i.e., N_{sp} from $\approx 7.4 \times 10^7$ to $\approx 1.4 \times 10^8$) and ≈ 1.6 times (i.e., E_{sp} from ≈ 3.3 to $\approx 5.2 \text{ pJ}$), respectively. The evident increase in E_{sp} and N_{sp} can be attributed to the higher field energy confined in a flat slit than that in a line slit (for reference, $\approx 32\%$ for the flat slit versus $\approx 16\%$ for the line slit).

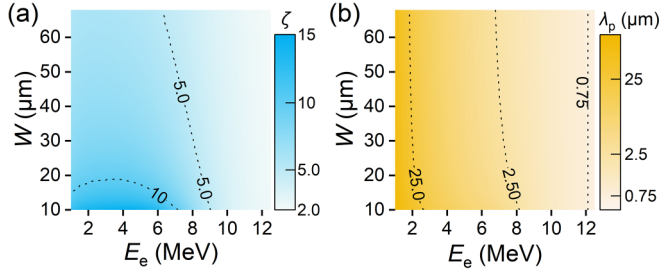


FIG. 3. Dependence of (a) ζ and (b) λ_p on W and E_e , with $\lambda_0 = 750 \mu\text{m}$ and electron density $8.6 \times 10^{23}/\text{m}^3$.

Since the profile of the sub-half-cycle pulse shown in Fig. 2(a) differs from that of a conventional Gaussian profile, we define the effective energy (E_{eff}) herein as the fractional energy of the pulse contained within the temporal slot of the pulse width τ as $E_{\text{eff}} = \int_{-\tau/2}^{\tau/2} I(t) dt$, for evaluating the available pulse energy with sub-half-cycle feature. It shows that, compared with $E_{\text{eff}} \approx 1.8 \text{ pJ}$ (i.e., 55% of 3.3 pJ) of a sub-half-cycle pulse generated with the previous line-slit-confined optical field, here E_{eff} increases to $\approx 4.0 \text{ pJ}$ (i.e., $\approx 77\%$ of 5.2 pJ) with the flat-slit-confined optical field, indicating a higher effective energy and higher ratio of effective energy out of the total pulse energy.

With fixed parameters of the driving field and the individual sapphire square waveguide (with $d = 125 \mu\text{m}$), the temporal confinement factor ζ and the peak wavelength λ_p of the ICS pulse can be changed by changing the electron energy E_e and the slit width W (here, the short-axis width of the electron bunch is assumed to be W). Figures 3(a) and 3(b) show the dependence of ζ and λ_p on E_e and W , respectively. It shows that ζ typically decreases with increasing W and E_e [Fig. 3(a)], and λ_p decreases with increasing W and E_e [Fig. 3(b)]. Therefore, the two important parameters, ζ and λ_p , of a sub-half-cycle pulse, can be selected within a wide range by changing E_e and W .

To obtain sub-half-cycle pulses with even shorter pulse width (e.g., attosecond scale), the pulse width of the electron bunch should be reduced to reduce the pulse broadening from the electron pulse, and higher electron energy and shorter pulse width should be used to reduce the time of electron bunch transiting through the flat-confined driving field with similar spatial FWHM. Considering recent advances in generating attosecond electron pulses [34–36], such ultrafast electron bunches may be accessible in the near future. Herein, to investigate the potential of generating an attosecond sub-half-cycle pulse, we use a 30-MeV 50-as 50-pC electron bunch (with an electron density of $8.5 \times 10^{23}/\text{m}^3$, the same as that in the previous work [20]), and the same driving field as in Fig. 2 (i.e., $\nu_0 = 0.4 \text{ THz}$, $\tau_0 = 1 \text{ ps}$, $d = 125 \mu\text{m}$, $W = 25 \mu\text{m}$) for ICS pulse generation. Figure 4 shows the calculated results of the attosecond ICS pulse, which has $\tau \approx 80 \text{ as}$ and $\nu_p \approx 1.7 \text{ PHz}$ (corresponding to $\lambda_p \approx 172 \text{ nm}$ and $T_c \approx 0.59 \text{ fs}$). Thus, the pulse has a temporal confinement factor $\zeta \approx 7.3$, corresponding to a 0.14-cycle 80-as ultraviolet attosecond pulse.

Meanwhile, the N_{sp} obtained from Eq. (2) is $\approx 7.0 \times 10^7$, with E_{sp} derived from Eq. (8) of $\approx 151 \text{ pJ}$, and $E_{\text{eff}} \approx 111 \text{ pJ}$ (74% of 151 pJ). Compared with that in Fig. 2, the much

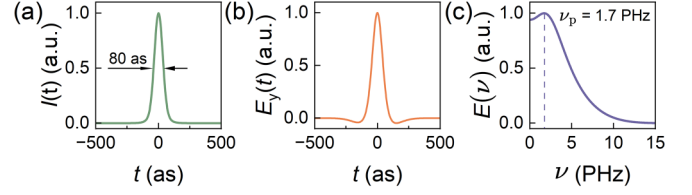


FIG. 4. Sub-half-cycle attosecond ICS pulse generated with a 0.4-THz 1-ps flat-slit-confined optical field (with Δs of $128 \mu\text{m}$) and a 30-MeV 50-as electron bunch. (a) Intensity $I(t)$, (b) electric field $E_y(t)$, and (c) Fourier spectrum $E(\nu)$ of $E_y(t)$ of the ICS pulse, giving $\tau \approx 80 \text{ as}$ and $\nu_p \approx 1.7 \text{ PHz}$.

higher single-pulse energy E_{sp} obtained here stems from the much higher average photon energy (for reference, ν_p increased from ≈ 0.022 to $\approx 1.7 \text{ PHz}$).

V. CONCLUSION

In summary, we have demonstrated an approach for generating sub-half-cycle ultrafast optical pulses via ICS in a flat-slit-confined optical field. The use of sapphire square waveguides to construct the flat slit can not only reduce the difficulty in fabricating the sub-half-wavelength field confinement structure, but also increase the single-pulse energy of the sub-half-cycle ICS pulse. Our calculation results show that, using ICS between a 1-fs 3-MeV 0.1-nC electron bunch and a 0.4-THz 1-ps flat-slit-confined optical driving field, a 0.12-cycle 5.1-fs-width midinfrared femtosecond pulse can be obtained. Compared with the line-slit approach, under similar conditions, here the flat-slit approach can increase the single-pulse energy by a factor of ≈ 1.6 (e.g., from ≈ 3.3 to $\approx 5.2 \text{ pJ}$), and the effective single-pulse energy by a factor of ≈ 2.2 (from ≈ 1.8 to $\approx 4.0 \text{ pJ}$). Also, by employing a 30-MeV 50-as 50-pC electron bunch and the same 0.4-THz optical driving field, it is possible to generate a 0.14-cycle 80-as-width ultraviolet attosecond pulse, with a single-pulse energy of 151 pJ and an effective single-pulse energy of 111 pJ. Overall, compared to the previous line-slit field confinement approach, the flat-slit configuration given here provides a more feasible approach for experimental realization of sub-half-cycle ICS pulses, with easier structure fabrication and higher single-pulse energy.

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DATA AVAILABILITY

The data that support the findings of this article are openly available [37].

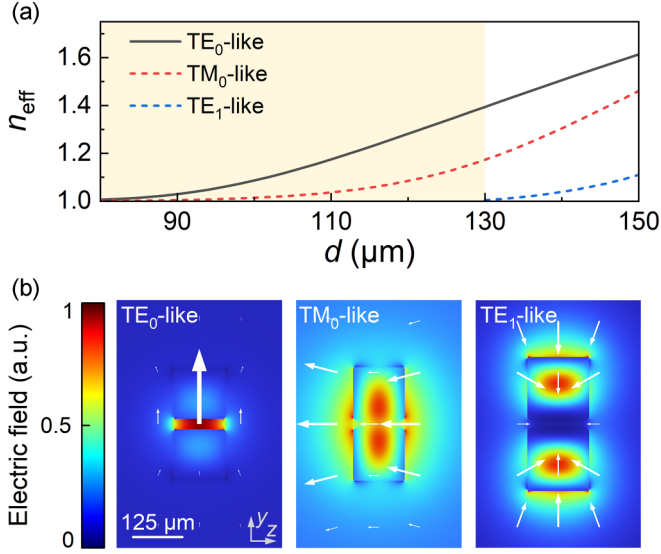


FIG. 5. Three lowest-order eigenmodes of a flat-slit-confinement structure excited with a 0.4-THz driving pulse. (a) Side length (d)-dependent effective refractive indices (n_{eff}) of the three lowest-order eigenmode. (b) Normalized electric-field distributions of TE₀-like, TM₀-like, and TE₁-like modes (from left to right). In the calculation, the side length d is assumed to be 125 μm for the first two modes (i.e., TE₀-like and TM₀-like modes) and 150 μm for the TE₁-like mode; the fillet radius of the edge is assumed to be 5 μm. The white arrows indicate the electric-field directions.

APPENDIX: DETAILS

In the Appendix, we provide additional information related to the main text above. Throughout the Appendix, we use the same notations used in the main text.

1. Waveguiding modes in a flat-slit-confinement structure

The flat-slit-confinement structure formed by the sapphire square waveguide ($d = 125 \mu\text{m}$, $W = 25 \mu\text{m}$) supports strong-coupling-induced hybrid eigenmodes under the optical driving pulse ($\lambda_0 = 750 \mu\text{m}$, $\nu_0 = 0.4 \text{ THz}$). The three lowest-order eigenmodes and their electric-field distributions are shown in Fig. 5.

Due to the broken-symmetry structure of the flat-slit-confinement structure, its fundamental mode splits into two

noncutoff modes with mutually perpendicular polarizations [38], namely the TE₀-like and TM₀-like modes. The higher-order TE₁-like mode is cut off for a side length $d < 130 \mu\text{m}$ (as shown in the light yellow region in Fig. 5). For generating sub-half-cycle pulses via ICS, the TE₀-like mode is required. Experimentally, the TE₀-like mode can be selectively excited by using a y-polarization THz driving pulse in a flat slit with $d < 130 \mu\text{m}$.

For the TE₀-like mode, the hot surface distributes between the two interfaces of the flat slit, which may induce atomic ionization of the material and sample damage when the field intensity is too high [23]. In the calculations herein, the peak power of the driving pulse P_{pulse} in the slit is assumed to equal the damage threshold of sapphire DT , with the calculation formula given as

$$\frac{DT}{P_{\text{pulse}}} = \frac{S_{\text{HS}}}{P_{\text{mode}}}, \quad (\text{A1})$$

where $DT = 2 \times 10^{18} \text{ W/m}^2$ is the damage threshold of single-crystal sapphire [23], S_{HS} is the peak x-Poynting vector on the hot surface, and P_{mode} is the total power of the mode (i.e., TE₀-like).

2. Electron trajectory in the flat slit

We employed a classical relativistic electrodynamics method to calculate the trajectory and velocity of electrons in the flat slit. Specifically, due to the high energy of the electrons (i.e., 3 MeV) assumed in this work, the space-charge effect within the electron bunch can be neglected [39], and the trajectory of an electron in the optical driving field can be used to approximate the path of the electron bunch [40].

Within the classical relativistic electrodynamics, the ICS is regarded as a process in which electrons deflect and radiate secondary waves under the electric and magnetic fields of the driving light. The relativistic kinematic equation is given by

$$\frac{d(\gamma\beta)}{dt} = \frac{e(\mathbf{E} + \mathbf{V}_e \times \mathbf{B})}{m_0 c}, \quad (\text{A2})$$

where $e = 1.603 \times 10^{-19} \text{ C}$ is the electron charge, $m_0 = 9.11 \times 10^{-31} \text{ kg}$ is the electron rest mass, and $\mathbf{E}$ and $\mathbf{B}$ are the electric- and magnetic-field vectors, respectively. Using Eq. (A2) and the finite difference time domain simulation, the velocity magnitude and trajectory of the electron's motion can

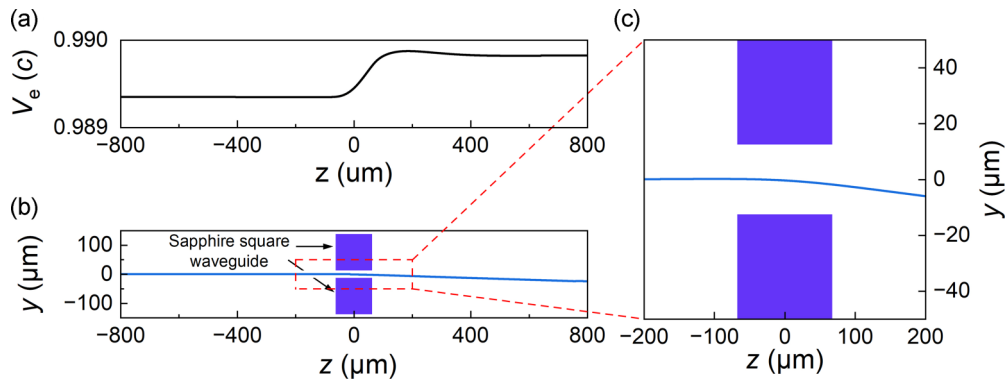


FIG. 6. Electron velocity and trajectory in the flat-slit-confined THz driving field. (a) z -dependent electron velocity V_e . (b) Electron trajectory in the y - z plane. (c) Close-up view of the electron trajectory with enlarged y -axis displacement.

be obtained. As shown in Fig. 6, electron velocity V_e changes only $\approx 0.06\%$ after passing through the slit [Fig. 6(a)], and the

electron trajectory can be approximated as a straight line when passing through the slit [Figs. 6(b) and 6(c)].

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